

PROJECT DEVELOPMENT

1.Introduction

The Project Development phase focuses on implementing the ComicCraft application based on the requirements and design prepared in the previous phases. The application is developed as a web-based AI comic story creator that converts a user's story idea into a comic-style output.

The system combines a FastAPI backend, HTML/CSS/Jinja2 frontend, Google Gemini models, image-generation technology, and PDF export functionality.

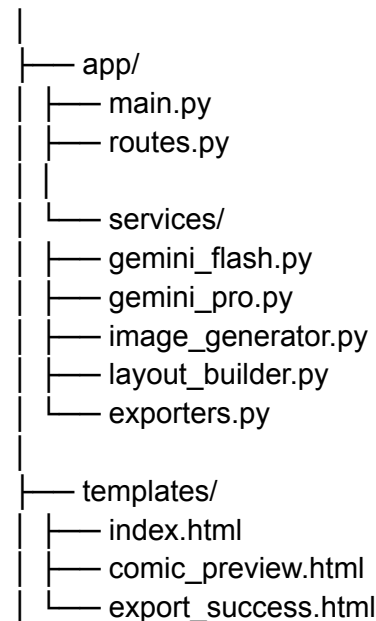
2.Technologies Used

- Frontend: HTML, CSS, Jinja2 Templates
- Backend: Python, FastAPI
- AI Story Generation: Google Gemini API
- Image Generation: AI image-generation model/service
- PDF Generation: FPDF
- Development Environment: Visual Studio Code
- Server: Uvicorn
- Version Control: GitHub

3.Development Structure

The project was organized into separate files and folders to make development and maintenance easier.

ComicCraftAI/



|
└─ requirements.txt

4.Backend Development

The backend was developed using Python and FastAPI.

The main backend components handle:

- Receiving comic details from the user
- Processing the story prompt
- Communicating with the Gemini API
- Generating comic content
- Handling generated images
- Preparing the comic preview
- Exporting the final comic as a PDF

The FastAPI server is executed using Uvicorn.

5.Frontend Development

The user interface was developed using HTML, CSS, and Jinja2 templates.

The main input page allows the user to provide:

- Story prompt
- Main character name
- Setting
- Tone
- Art style

The generated comic is then displayed on the comic preview page.

6.AI Integration

Google Gemini is integrated into the application to generate the comic story content.

The AI processing is used to create:

1. A comic story outline
2. Panel-wise narration
3. Character dialogue
4. Comic-related content based on the user's prompt

The generated information is passed to the application for displaying the comic panels.

7. Image Generation

The project includes an image-generation component for creating visual content for the comic panels.

The image generator receives information such as:

- Character
- Scene
- Setting
- Art style
- Story context

The generated images are then used in the comic preview.

8. Comic Preview Development

The Comic Preview page combines the generated story content and images into a panel-based comic format.

Each panel can contain:

- Panel image
- Narration
- Dialogue
- Character information

This allows the user to view the generated comic before exporting it.

9. PDF Export Development

The application includes a PDF export feature using FPDF.

The generated comic content is arranged into a PDF document containing the comic panels, images, narration, and dialogue.

The user can download the completed comic as a PDF.

Application Execution

The application can be started using the Uvicorn development server.

The development server runs the FastAPI application locally, allowing the developers to test the application's pages and features through a web browser.

The application was tested through the local server to verify the main workflow:

User Input → AI Story Generation → Image Generation → Comic Preview → PDF Export

11.Development Outcome

The development phase resulted in a functional prototype of ComicCraft. The application provides an integrated workflow for creating AI-assisted comic stories from a simple user prompt.

The major developed components include:

- User input interface
- FastAPI backend
- Gemini AI integration
- Image-generation component
- Comic preview interface
- PDF export functionality

12.Conclusion

The Project Development phase transformed the planned design of ComicCraft into an implemented web application. The different components were integrated to provide a simple workflow for generating comic stories using AI.

The developed application was then prepared for the Project Testing phase, where its functionality and output were verified using different test cases.